Assignment 17 — Solutions

Project & Behavioural Questions

Part D — Interview Scenarios · 6 questions with answers

Q1. Walk me through a GenAI project you built end-to-end. What was your role and what was the business impact? ★

Answer

How to answer (STAR + architecture, ~3 minutes):

- **Situation/Problem:** business context and why it mattered. "Fraud analysts spent ~20 minutes per case searching policy documents and past case notes."
- **Your role:** what *you* owned (design, RAG pipeline, evaluation, deployment) and team size.
- **Solution/architecture:** walk through data → ingestion/chunking → embeddings → vector DB → retrieval (hybrid + re-ranker) → LLM → guardrails → UI/API → monitoring. Name the tools and **why** you chose them.
- **Evaluation:** how you measured quality (golden set, RAGAS faithfulness, accuracy) and the numbers.
- **Result/impact:** quantified – "reduced investigation time from 20 to 6 minutes, used by 150 analysts, 85% answer acceptance".

Tips: prepare one project in depth with a diagram you can draw; be honest about what was a prototype vs production; expect follow-up questions on every component.

Q2. What was the hardest problem you faced in that project and how did you solve it? ★

Answer

Structure: challenge → why it was hard → options you considered → what you did → measurable result → lesson.

Example answer: "Our RAG bot gave wrong answers for questions about tables in scanned PDFs. Plain text extraction scrambled the rows, so retrieval found the right page but the LLM couldn't read the numbers. I compared three approaches – better OCR, a vision-LLM per page, and a table-aware parser – on 50 table questions. A layout-aware parser that converted tables to Markdown, kept each table as one chunk with an LLM-generated summary for retrieval, raised accuracy on table questions from 41% to 83%. Lesson: evaluate each component separately and fix the data pipeline before changing the model."

Pick a genuinely technical problem (data quality, hallucination, latency, cost, evaluation), show your reasoning and trade-offs, and quantify the outcome.

Q3. How did you evaluate your system and what metrics improved?

Answer

Interviewers want to see a **disciplined evaluation process**, not just "it worked well":

- **Test set:** "We built a golden set of 150 real questions with expert-written answers and source documents, including out-of-scope questions."
- **Metrics by component:** retrieval – hit rate@5, context recall/precision; generation – faithfulness, answer relevancy/correctness (RAGAS + LLM-as-judge validated against human

labels); system – latency p95, cost per query.

- **Baselines and iterations:** "Baseline top-5 vector search: 68% hit rate → adding BM25 hybrid: 79% → adding a re-ranker: 88%; faithfulness improved from 0.81 to 0.93 after prompt changes."
- **Online metrics:** thumbs-up rate, escalation rate, adoption, time saved.
- **Continuous evaluation:** regression tests in CI on every prompt/model change; production failures added to the test set.

Q4. What would you do differently if you rebuilt it today?

Answer

Show **reflection and awareness of newer techniques**; be honest but positive. Example points:

- "Build the **evaluation set on day one** – we added it late and wasted time on changes we couldn't measure."
- "Use **hybrid search and a re-ranker** from the start instead of pure vector search."
- "Invest earlier in **document parsing** (tables, scanned PDFs), which caused most errors."
- "Add **observability/tracing** (Langfuse/LangSmith) before launch."
- "Use **LangGraph** for explicit control instead of a loosely controlled agent; add human-in-the-loop for risky actions."
- "Consider **smaller models + routing** and prompt caching to reduce cost."
- "Involve end users earlier to collect realistic queries."

Avoid blaming others; focus on what you learned.

Q5. How do you keep up with the fast-changing GenAI field?

Answer

Give **specific, credible habits** (and mention something recent you actually learned):

- **Papers & research:** arXiv / Hugging Face Daily Papers, Papers with Code; reading key papers (Attention Is All You Need, LoRA, RAG, ReAct, DPO).
- **Official docs and engineering blogs:** OpenAI, Anthropic, Google DeepMind, Meta AI, Hugging Face, LangChain/LlamaIndex blogs and release notes.
- **Newsletters/communities:** The Batch (DeepLearning.AI), Latent Space, Ahead of AI (Sebastian Raschka), Interconnects, r/LocalLLaMA, X/LinkedIn researchers, local meetups.
- **Courses:** DeepLearning.AI short courses, Hugging Face courses.
- **Hands-on:** build small projects to test new models/techniques (e.g. tried MCP servers, GraphRAG, a new open model), Kaggle, open-source contributions.
- **Teaching/sharing:** writing blog posts or teaching classes forces deep understanding.

Q6. Explain RAG (or a Transformer) to a non-technical manager in 2 minutes.

Answer

RAG explained simply (example):

"An AI model like ChatGPT is like a very smart new employee who has read a lot of the internet, but has **never seen our company's documents** and sometimes **guesses confidently** when unsure.

RAG gives this employee an **open-book exam**. When someone asks a question, the system first **searches our own documents** – policies, manuals, past cases – and picks the few most relevant pages. It then gives those pages to the AI and says: 'Answer **only** using these pages, and tell us which page you used.'

Benefits: answers are based on our latest, approved information; every answer shows its source so people can verify it; when a policy changes we just update the documents – no expensive retraining; and sensitive documents can be restricted to the right people.

Limitations: it's only as good as the documents and the search – if the information isn't there, it should say 'I don't know'."

Tips: use an analogy, avoid jargon (embeddings, vectors), focus on business value, risks and cost; check for understanding.

(Transformer in one line for managers: "a model that reads all words of a sentence at once and learns which words to pay attention to, which is what makes modern AI understand context so well.")